Using LLMs in research

(in September 2026)

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Six things to keep in mind

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6. **Think in workflows, not tools**

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- I worry much less about fake citations than I used to

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Experiment with these models

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- Claude Code or Codex instead of a chat window
- Spend time on your `CLAUDE.md` or `AGENTS.md` (standing instructions for every session)

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- Recent math: Erdős problems, a Millennium problem (disputed) (Quanta, Aug 2026) (OpenAI, Sep 2026)
- We pretend at our peril that these models can’t do a lot of great things in our research

A couple of my favorite things

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- Data visualization

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- Have several agents do the task and check it, then verify it yourself

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- But don't be stupid

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Everyone else has AI too

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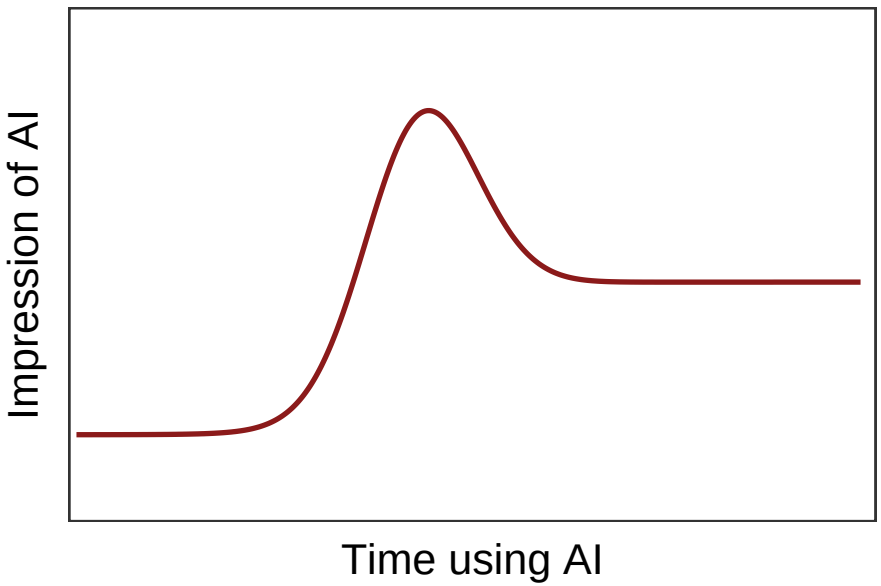
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- Ideally, use AI to raise the ceiling rather than the floor

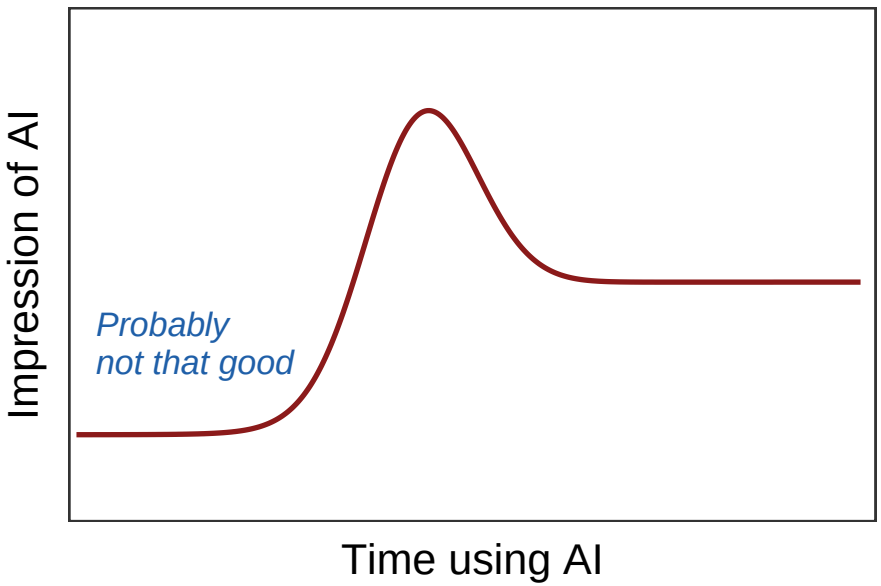
6

Think in workflows, not tools

Impressions change with use

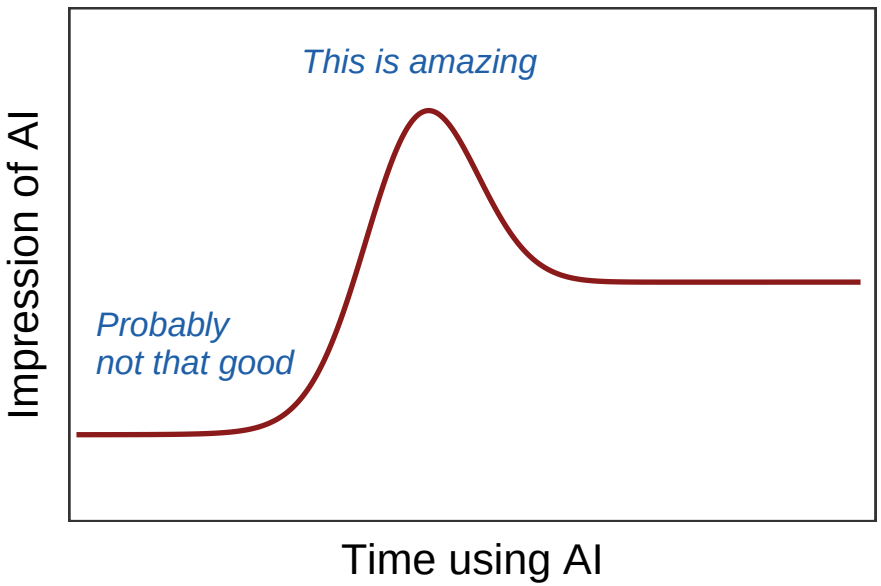


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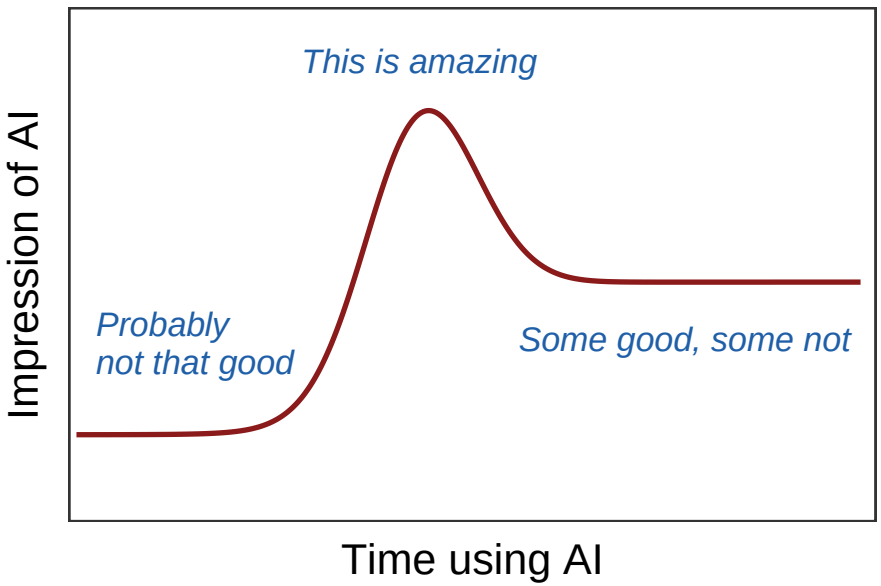
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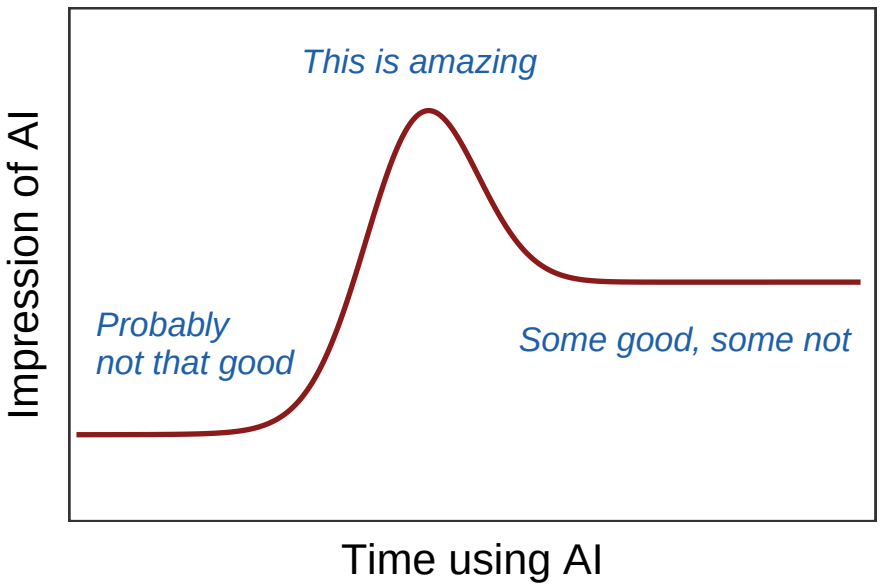
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- Also addictive, like social media

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- No AI: careful writing, often by hand, on paper, away from the computer
- Heavy AI: kick off a bunch of agents to explore data and try new things

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